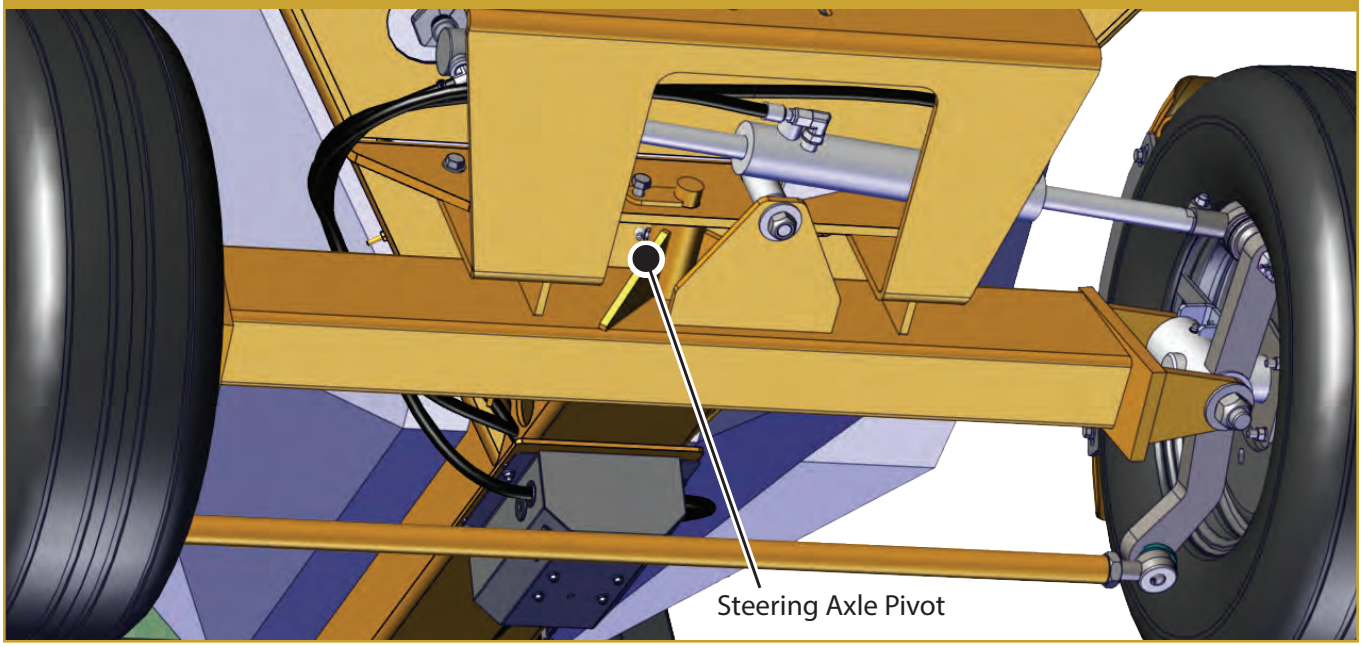
Monthly Lubrication



Biannual and Annual Lubrication

Drive Axle

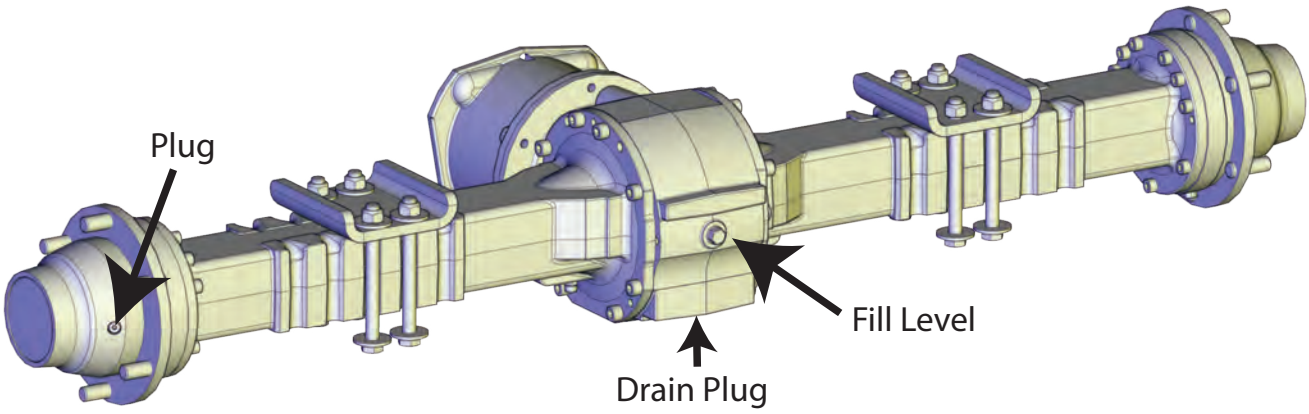
Check the drive axle oil level at least every six months and fill with gear oil if needed.

To fill the oil:

1. Rotate the hub on one side of the front axle until the plug is in the 12 o'clock position.
2. Remove the plug.
3. Add one quart of 80W-90LS GL5 oil..
4. Replace the plug.
5. Follow Steps 1 - 4 on the other side.

Change the drive axle oil every 12 months:

1. Remove the level plug in the center of the housing.
2. Remove the breather plug in the housing.
3. Add oil until it starts coming out of the level plug.
4. Replace the level plug.
5. Allow the axle to sit level for at least five minutes to allow the oil to flow to the hubs.
6. Remove the level plug and check oil level.
7. Add more oil if needed.



ENGINE MAINTENANCE

The Hatz 74 HP is a turbo-charged, four-cylinder diesel engine that also drives the hydraulic auxiliary pumps for steering and broom control. The engine performs at 2500 RPMs with peak torque at 184 ft lb @ 1600 RPMs and displacement at 119 cu in.

The diesel engine manufacturer recommends several engine-specific maintenance inspections every 250 hours, or every three months. You can access the engine and other machine components via the access panel doors on either side of the machine, depending upon the component. **(Figure 5-3)** (Refer to your Engine Operator's Manual for more detailed information regarding engine-related maintenance procedures.)

Check Engine Oil

The engine lubrication oil must be kept at a level above the ADD mark, but not above the FULL mark, on the engine lubrication oil dipstick.

To accurately check the engine lubrication oil level, park the machine on a level surface and stop the engine.

⚠WARNING ALWAYS stop the engine before checking the engine lubrication oil level as hot oil can be thrown, causing serious injury.

NOTICE Wait five minutes after engine shutdown before removing the dipstick from the engine and checking the oil level to assure accuracy. This also prevents filling the engine with too much lubrication oil. Engine oil should be warm when checked.

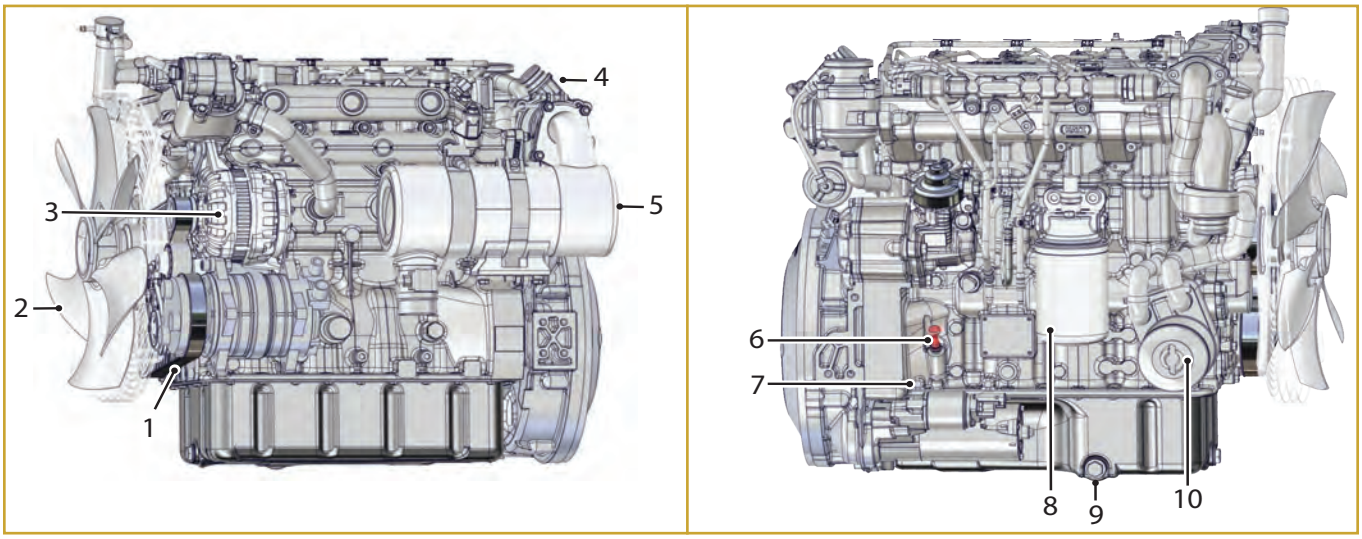


Figure 5-3. Engine Components

1 - Engine Belt

2 - Engine Fan

3 - Alternator

4 - Exhaust Gas Turbocharger

5 - Diesel Oxidation Catalytic Converter (DOC)

6 - Engine Oil Dipstick

7 - Bottom Engine Oil Filler Plug

8 - Fuel Filter

9 - Engine Oil Drain Screw

10 - Oil Filter

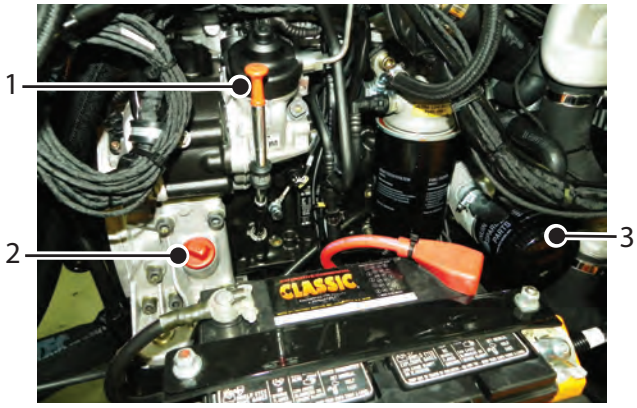


Figure 5-4. Engine Oil Components

1 - Oil Dipstick

2 - Oil Filler Cap

3 - Oil Filter

1. Open left engine access panel to access the oil dipstick. **(Figure 5-4)**
2. Clean around the engine lubrication oil dipstick, then remove it from the engine.
3. Wipe the dipstick with a clean cloth, then reinsert.
4. Pull dipstick out of the engine and check that oil level is near the full mark on the dipstick.
5. To add oil, remove the filler cap located next to the dipstick. Twist to remove cap and twist to install (use 15W40 motor oil).

Change Engine Oil and Filter

⚠WARNING **DO NOT change the engine lubrication oil when the engine and lubrication oil are hot. Hot oil can cause serious injury.**

NOTICE **DO NOT change the engine lubrication oil while the engine is running as serious damage or engine failure can occur.**

NOTICE **The engine lubrication oil filter must be changed when the engine lubrication oil is changed.**

While the engine is warm, proceed as follows:

1. Clean the area around the engine oil dipstick.
2. Place a container that has sufficient capacity to hold the drained oil directly under the engine oil drain plug.
3. Carefully remove the engine lubrication oil drain plug to allow the oil to drain into the container.
4. Wipe the area around the engine lubrication oil filter element and its mounting base with a clean cloth. **(Figure 5-4)**
5. Place the container under the filter element.
6. Use a filter removal wrench to loosen and remove the filter element by turning it in a counterclockwise direction.
7. Drain and discard the removed filter element.

NOTE: Be sure the used rubber gasket is removed and discarded with the filter element.

NOTE: Comply with environmental laws by properly discarding used oil, filter and gasket.

8. Wipe the inside area of the lubrication oil filter head using a clean, lint-free cloth.
9. Rub a little oil on the rubber gasket area of the new filter element and fill the new filter element with the oil.
10. Install the new filter element onto the filter head.
11. Carefully tighten the filter element (by hand only).

NOTE: Tighten the filter element as directed by the filter manufacturer.

12. Clean, reinstall and carefully tighten the lubrication oil drain plug.

NOTICE **DO NOT overtighten the drain plug.**

13. Fill the engine with 7.4 quarts of oil.
14. Reinsert the dipstick.

Inspect Cooling System

1. Open the right engine panel door to access the cooling fan. **(Figure 5-5)**
2. Check engine cooling fan for damaged or bent blades. Replace if damaged.
3. Inspect the fan belt for serviceability, looking for signs of cracking, tearing or separation. Replace if damaged.
4. Check fan belt and pulleys for dust, dirt and debris. Clean before starting the engine.

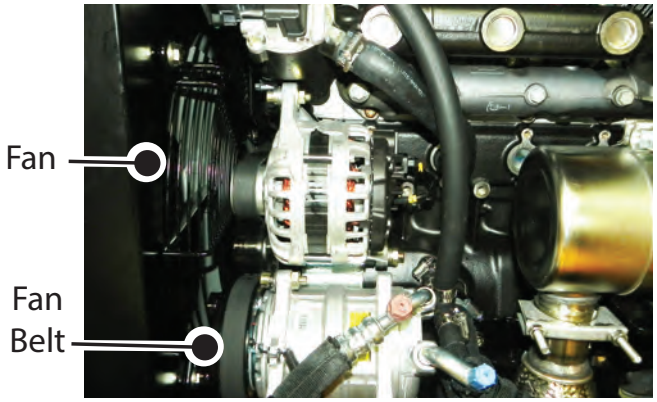


Figure 5-5. Fan and Fan Belt

Air Filter

The heavy-duty engine air filter system is mounted inside the (left) engine compartment. **(Figure 5-6)** To prolong engine life and prevent contaminants from entering the engine, inspect the air intake system components daily for damage, cracked hoses or loose clamps.

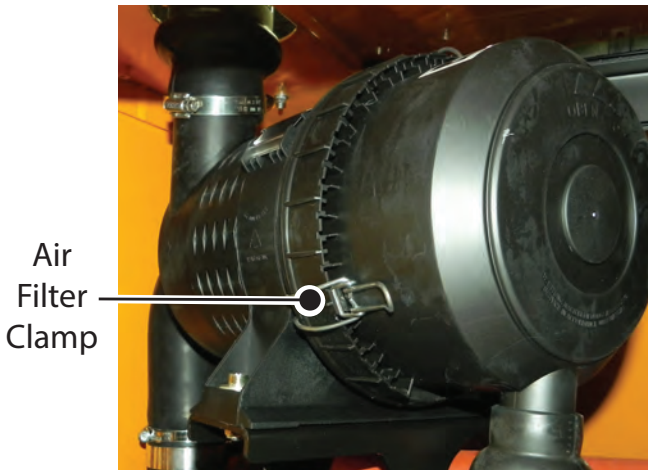


Figure 5-6. Air Filter and Fuel Filter

The air filter must be cleaned or replaced as follows:

1. Open clamps securing the cover on the filter housing, then remove the cover.
2. Remove the filter cartridge.
3. Wipe inside the air filter housing with a damp cloth.
4. Replace with the new filter cartridge if needed.
5. Reinstall cover and secure with the clamps.

NOTICE **Never operate the engine without an air cleaner as this will cause severe engine damage.**

FUEL SYSTEM

Fuel Tank

Always fill the fuel tank at the end of each work day to reduce the possibility of moisture accumulation from condensation in the tank.

⚠WARNING NO SMOKING while filling the fuel tank as all fuels for internal combustion engines are flammable. Fill the fuel tank only in a designated area where there is good ventilation and a fire extinguisher available.

⚠WARNING NEVER overfill the tank, check fuel level or check for fuel leaks near an open flame or near equipment that can create sparks.

⚠WARNING Diesel fuel is very flammable. Use extra caution and DO NOT spill fuel.

Fuel Filter

The fuel filter elements are located inside the left engine access panel. **(Figure 5-7)**

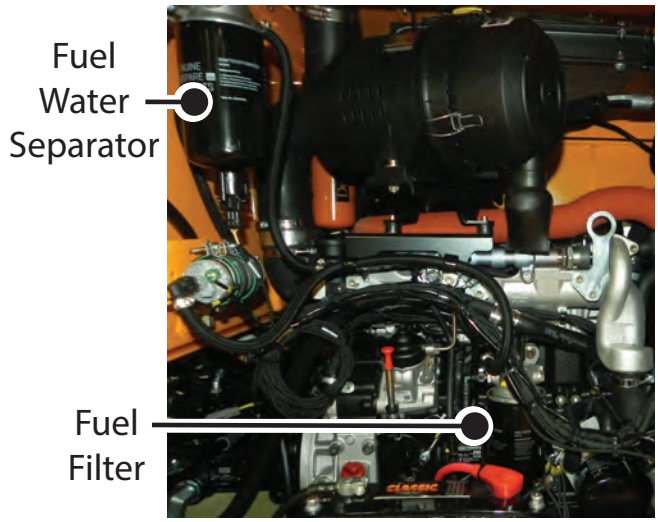


Figure 5-7. Fuel Filter and Water Separator Location

⚠WARNING DO NOT change the fuel filter while the machine is running. DO NOT change the fuel filter in an area near an open flame. DO NOT smoke while changing the fuel filter.

⚠DANGER NEVER loosen a fuel injector line as contents are under high pressure. DO NOT try to bleed air by loosening injector lines. NEVER loosen the bleed screw on the fuel filter. Large quantities of fuel under high pressure can escape and cause serious injury.

Replace the fuel filter using the following general procedures:

1. Stop the engine.
2. Put a container under fuel filters before removing the filter elements.

NOTE: Consider the environment when discarding used filters. Dispose of fuel filters in accordance with safe and lawful practices.

3. Wipe the area around the fuel filter elements and element mounting heads using a clean, lint-free cloth.
4. Use a filter removal wrench to loosen and remove fuel filter elements by turning in a counterclockwise direction. Drain and discard the removed elements.
5. Wipe the inside area of the filter heads with a clean, lint-free cloth. Fill the new fuel filter elements completely with clean fuel.
6. Put clean fuel onto the element rubber gaskets.
7. Install the new fuel filter elements onto the filter heads. Carefully tighten the elements (by hand only). **DO NOT overtighten the fuel filter elements onto the filter heads.**

NOTE: After changing the fuel filter, the ignition key must be set to the ON position for a few seconds sot he feed pump fills the filter with fuel and air is pushed out of the system.

8. Start the engine and check for ANY fuel leaks.

⚠WARNING STOP the engine immediately if any fuel leakage is noted. DO NOT start the engine until the leakage problem is corrected.

Fuel Water Separator

If water gets into the fuel system, it settles into the water separator. **(Figure 5-8)** When the water level reaches a certain level, an error code will appear on the PV380 digital display. It is important to drain it immediately.

NOTICE **If you run out of fuel, it is important to pump the water separator to prime fuel injection lines. Failure to do so can adversely affect performance.**

To drain the water separator:

1. Pull retainer clip from plug to disconnect the harness.
2. Unplug the harness, then unscrew the plug.
3. Drain and reinsert components.



Figure 5-8. Fuel Water Separator

To clean the water separator:

1. Follow the previous steps for draining the water separator.
2. Loosen and remove the filter body and clean it with light oil. Replace if needed.
3. Reinsert and tighten.

ELECTRICAL SYSTEM

Battery Servicing

The electrical system is powered by a 12-volt,negative-ground, maintenance-free battery located in its own compartment on the side of the cab. The master battery On/Off switch is also located in this compartment. **(Figure 5-9)**

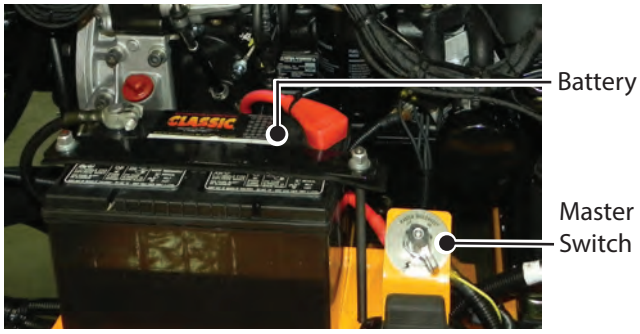


Figure 5-9. Battery

NOTE: The electrical system is a negative-ground system. Connect the positive (+) cable to the positive (+) post of the battery. Connect the ground cable to the negative (-) post of the battery. It is advisable to disconnect the negative (-) cable first and reconnect it last. Reversed polarity can damage the electrical system.

Clean terminals and battery posts thoroughly and check that battery cable terminals are tight. Dirty or loose connections can create high electrical resistance and permit arcing, which will quickly burn and pit terminals and posts.

Keep the battery clean by washing it off whenever dirt buildup is excessive. If corrosion is present around terminal connections, remove them and wash with ammonia solution or a solution consisting of 1/4 lb. (11 kg) baking soda added to one quart of warm water. Make certain the vent caps are tight to prevent solution from entering the cells. After cleaning, pour clean water over the battery and surrounding area to wash the solution away. Check vent cap breather openings to make sure they are open.

⚠WARNING Batteries contain sulfuric acid and normally produce explosive gases that can cause serious injury. DO NOT allow flames or sparks to come near the battery.

⚠WARNING When charging or working near a battery, always shield your face and wear eye protection. In case of acid contact with skin or eyes, flush immediately with water for a minimum of 15 minutes and get prompt medical attention. If acid is swallowed, call a physician immediately.

⚠CAUTION When servicing the electrical system or welding on the machine, always turn the Master Disconnect switch to the OFF position to disconnect the ground strap from the battery to prevent damage to the machine electrical system.

If the battery repeatedly discharges or the engine is difficult to start, check the electrical charging system. Using a battery tester, you can check the voltage following these procedures:

1. Shut down the engine and remove ignition key.
2. Place the positive (red) multimeter lead on the positive (+) battery terminal; and the negative (black) multimeter lead on the grounded negative (-) battery terminal.
3. With the engine off and the multimeter set at 12 VDC, the battery must show a charge of 12 - 12.5 volts.
4. Start the engine.
5. Check the battery charge level. The multimeter should read at least 13.5 volts.

Also check that the circuit breaker has not tripped. The circuit breaker is located inside the engine compartment on the auxiliary panel. **(Figure 5-10)**

If the battery does not charge properly, you may need a new battery or the alternator may need repair or replacement.

If storing the machine for more than 30 days, be sure to remove the battery and store it separately in a cool, dry place. Continue to keep the battery fully charged and check the level of electrolyte regularly to prevent damage or corrosion.

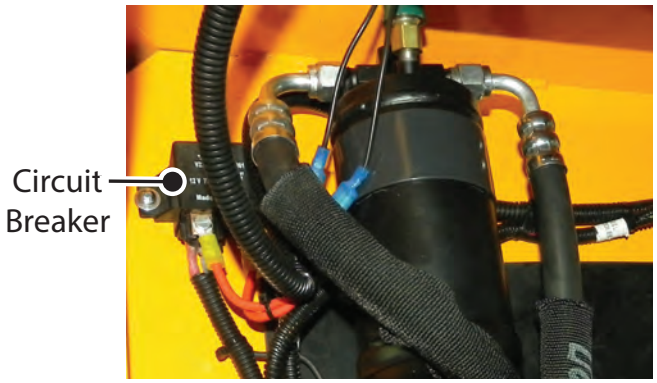


Figure 5-10. Circuit Breaker

Check Battery Electrolyte Level

⚠WARNING Batteries create explosive gases when charging. DO NOT expose these gases to sparks or open flame during maintenance. Always perform battery service in a well-ventilated area or outside. Serious injury can occur from exploding batteries.

⚠CAUTION Do not attempt to charge a frozen battery. Allow a frozen battery to thaw completely before charging. If using serviceable batteries, fill each cell to specified range with distilled water. DO NOT overfill.

6. Check battery charge and charging rate.
7. Shut down machine by turning ignition switch key counterclockwise to the OFF position.
8. Using a multimeter, check the battery charge by placing the positive (red) multimeter lead on the positive (+) battery terminal, and the negative (black) multimeter lead on the grounded (-) battery terminal. With the multimeter set at the 12 VDC setting, the battery should show a charge of at least 12 volts. If not, charge the battery or perform a load test.
9. Start the engine. From the operator's seat, check the battery charge level. The multimeter should read at least 13.5 volts. If the multimeter does not indicate minimum charge, check the machine charging system.

⚠WARNING DO NOT dry batteries using compressed air. Allow the batteries and surrounding area to dry. Battery acid can cause severe injury if blown into the face or other areas of the body. If injury occurs, seek medical attention immediately. Flush eyes and skin for 30 minutes with clean water.

Alternator

The alternator supplies electrical power to electronic controls and charges the battery. Its built-in regulator controls voltage output. **(Figure 5-11)**

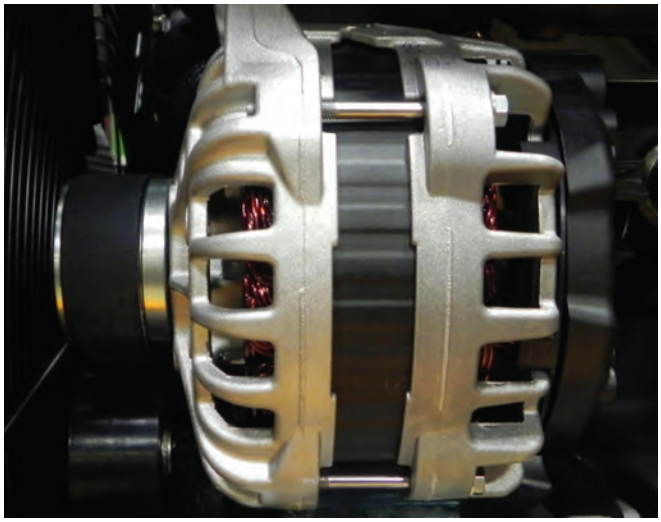


Figure 5-11. Alternator

Use the following precautions to prevent damage to the alternator or regulator:

- Never polarize the alternator.
- Never ground any alternator terminals or circuits.
- Always disconnect the battery before disconnecting or connecting the alternator.
- Never disconnect the alternator while machine is operating.
- Ensure wiring is properly connected before connecting the battery.
- Always connect a booster battery in the proper polarity: negative (-) to negative (-) and positive (+) to positive (+).

Lighting

Inspect lights for proper operation daily. If a light or group of lights does not function:

- Check the fuse box panel for a blown fuse.
- Examine all visible wiring connections, ensuring they are not frayed or damaged, and securely fastened.
- Check lighting mounts for proper ground.
- Remove light lens or casing and check bulb. Replace if needed.
- Inspect wiring harness for damage (see Schematics in **Section 6**).

If broken wires are found, solder together and cover with a shrink-wrap type of plastic covering (preferred) or electrician's tape to prevent contamination of the solder joint by moisture.

If you make any repairs to the wiring harness on the broom, always replace or repair the protective loom covering the wiring to prevent future damage to the wiring harness. Examine the routing of the harness and ensure it is not subjected to excessive movement, which often causes broken wiring.

Cold Weather Starting

NOTICE **Be sure to keep the battery fully charged during cold weather to keep it from freezing. Freezing weather has little effect on a fully charged battery.**

If a booster battery is necessary for cold weather starting:

1. Connect one end of the first jumper cable to the positive (+) terminal of the dead battery and the other end to the positive (+) terminal of the booster battery.
2. Connect one end of the second jumper cable to the negative (-) terminal of the booster battery and the other end to the frame of the machine with the dead battery.

Fuse Box Panel

The electrical system is protected from overload damage by fuses. If an electrical component fails to operate, check the fuse panel first to ensure that a fuse has not "blown." The panel is located on the right side panel in the cab. See the following table for corresponding fuse information. **(Figure 5-12)**

A blown fuse is an indication that there is an overload that needs to be corrected. If you replace a fuse, but it continues to blow, contact your authorized Rosco/LeeBoy dealer.

NOTICE **Always replace a blown fuse with the same rating as specified. Never replace with a higher amperage rating. Damage to the electrical system or fire can occur from improper fuse replacement.**

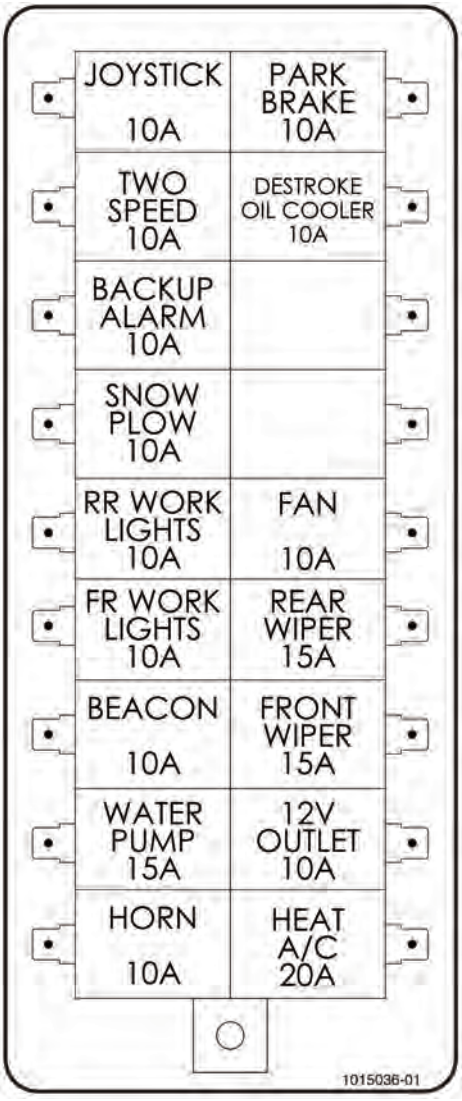


Figure 5-12. Fuse Panel and Fuse Sizes

Dielectric Grease

Dielectric grease is a nonconductive, insulating compound applied to electrical connectors. When applied correctly, a high-quality dielectric grease displaces debris and moisture in the connector. The proper application of dielectric grease fills any gaps between pins or in the connector which prevents debris and moisture from creating corrosion of the electrical components.

⚠CAUTION **Equipment damage can occur if using low-quality, incompatible dielectric grease.**

If not applied or incorrectly applied, corrosion can result that causes rapid increase in resistance, intermittent connections, overheating, and a variety of electrical problems. Numerous electrical issues throughout the machine can be avoided by the proper application and use of dielectric grease.



Figure 5-13. Correct Application of Dielectric Grease

To correctly apply dielectric grease:

- Apply a thin, visible, uniform coat using **ONLY** high-quality dielectric grease (Dow Corning 4 or equivalent) across the **MALE** connector as shown in **Figure 5-13.**
    - Coating too thin may not fill the gaps between pins.
    - Coating too thick may prevent proper connection or force pins or seals out. **(Figure 5-14)**
- Plug in the male connector.

⚠CAUTION DO NOT force connector to lock if too much grease has been applied.

⚠CAUTION Excessive grease application does not allow a proper connection by preventing the plugs from locking into place, and can cause damage to the connector seals.



Figure 5-14. Incorrect Application of Dielectric Grease

HYDRAULIC SYSTEM

The RB50 Broom features a hydraulic system consisting of an engine-mounted gear pump, three hydraulic cylinders, hydraulic reservoir, filter and a four-station valve block with solenoid-operated valves. The hydraulic motors and the hydraulic cylinders use the same hydraulic oil reservoir and oil supply.

The hydraulic pump and motor generally require no regularly scheduled maintenance, but inspect components for leaks or damage, which could be an indicator of the need for service. See your authorized LeeBoy/Rosco dealer if these components need servicing.

You should inspect the machine for hydraulic leaks daily, and check to ensure that hose fittings are secure and tight weekly.

If leaking fluid is found, it is probably on the pressure side of the hydraulic system. Find and repair the leaking component before using the machine to prevent damage to hydraulic components.

Leaks on the suction side of the hydraulic system are more difficult to find. This condition is serious since air or dirt introduced into the hydraulic system causes rapid component wear and eventual failure. Some symptoms of suction leakage are:

- Foaming of hydraulic oil.
- Sluggish hydraulic system operation.
- Unusual noise in the hydraulic pump or motor.

If you suspect a suction leak, verify that all reservoir connectors and fittings are properly tightened. If the problem persists, replace the defective hose assembly or fitting.

Contact your authorized LeeBoy/Rosco dealer for assistance and service repair if needed.

Your machine was factory-filled with Citgo All-Weather All-Temp VG32 hydraulic oil, a multi-grade anti-wear oil for use in equipment where wide temperature ranges are encountered. It is recommended that you use this hydraulic oil, or a comparable brand. Only introduce another brand after completely draining and flushing the hydraulic system, ensuring all hoses, cylinders, valves, motors and pumps have been purged of the factory-filled oil, and changing the oil filters.

⚠CAUTION Using other hydraulic oils could result in substandard performance or failure of the machine's hydraulic components.

⚠CAUTION DO NOT mix manufacturer or grade hydraulic oil weights. Substandard performance or hydraulic component failure can occur.

⚠WARNING NEVER attempt to repair hydraulic hoses with tape, clamps or cements. The hydraulic system operates under extremely high pressure. Temporary repairs will fail, creating a hazardous condition that can result in bodily injury and damage to the machine.

NOTICE Hydraulic oil that has oxidized or contains contamination can shorten the expected service life of the components in the hydraulic system.

Check Hydraulic Oil Level

Wait 10 minutes after engine has been shut down before checking hydraulic oil. If level is low, remove hydraulic oil fill cap strainer to add hydraulic oil to the tank.

Oil level is determined by sight gauge. **(Figure 5-15)** Hydraulic oil should be added if the oil level falls below the fill line. Never fill above the black fill line.

⚠CAUTION Use extreme caution when removing the filler cap to prevent any foreign matter from entering the hydraulic reservoir. This can cause damage to hydraulic components.

- Fill tank to the black fill line on sight gauge.
- Clean the fill cap strainer and reinstall it.

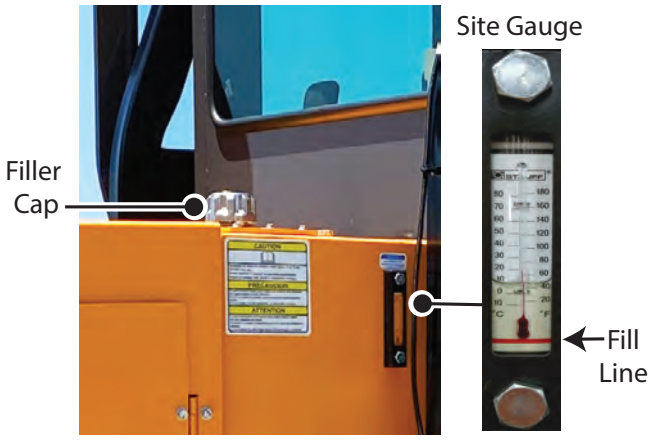


Figure 5-15. Hydraulic Sight Gauge and Filler Cap

Change Hydraulic Oil and Filter

Changing the hydraulic oil removes the accumulation of dirt, water and mechanical wear particles from the hydraulic oil reservoir and system. The chemical structure of hydraulic oil changes after continuous use, therefore, new oil is important to ensure correct operation of the hydraulic system.

NOTICE Never let the hydraulic oil reservoir tank run dry. Pump damage will occur.

Condensation can also build up in the hydraulic system and clog the filter elements. This could lead to insufficient hydraulic oil at the pump and degrade performance, causing damage to components.

Change the hydraulic oil and filter after the first 100 hours of use, then every 250 hours or seasonally, whichever comes first. Also remove and clean the suction strainer at the 500-hour intervals.

Use the following procedures to change the hydraulic oil and filter:

1. Stop the engine. Allow the hydraulic oil to cool until it is at a warm temperature. Slowly loosen and remove the hydraulic oil filler cap **(Figure 5-15)**. Put a clean, lint-free cloth over the reservoir fill tube opening and secure in place with tape.

⚠WARNING DO NOT drain the hydraulic oil from the reservoir when it is HOT. Hot hydraulic oil can cause serious injury. Drain at a warm temperature only.

2. Carefully remove the plug from the hydraulic tank. Use a drain collection container of sufficient capacity to collect the hydraulic oil underneath. Allow all of the hydraulic oil to drain into the container.
3. Check the hydraulic suction strainer in bottom of the hydraulic oil tank. Clean if necessary.
4. Install the hydraulic oil reservoir drain plug and tighten securely.

5. Unscrew hydraulic oil filter and remove. **(Figure 5-16)**

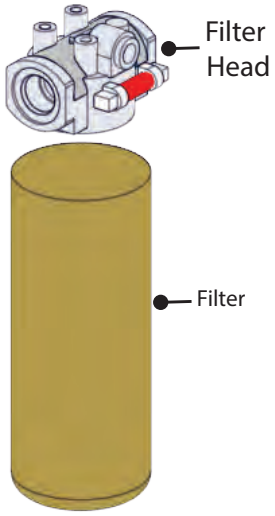


Figure 5-16. Hydraulic Oil Filter Components

6. Discard filter properly in accordance with government environmental regulations.
7. Reinstall new hydraulic oil filter.
8. Carefully remove the cloth from the hydraulic oil reservoir fill tube opening.
9. Refill the hydraulic oil reservoir (recommend Citgo All Weather/All Temperature Multiviscosity hydraulic oil).

NOTICE DO NOT fill the hydraulic oil reservoir with new hydraulic oil until the suction strainer has been cleaned. (See next section.)

NOTICE DO NOT overfill the hydraulic oil reservoir with oil.

10. Check the oil level at the sight gauge and add more if needed.
11. Replace the hydraulic oil fill strainer cap onto the reservoir filler neck and tighten securely.
12. Start the engine and operate hydraulic controls several times.
13. Check the hydraulic system for any leaks.

Hydraulic Pump and Motor

The hydraulic pump and motor generally require no regularly scheduled maintenance. Frequent inspection for leaks will indicate the need for service of these components. See your authorized LB Performance dealer for service if needed.

Change Hydraulic Oil Strainer

The oil strainer is mounted in the hydraulic oil filler opening under the filler cap. **(Figure 5-17)**

⚠WARNING DO NOT remove the hydraulic filler cap from the reservoir when it is hot as this can cause serious injury. Allow hydraulic oil to cool down to a warm temperature.

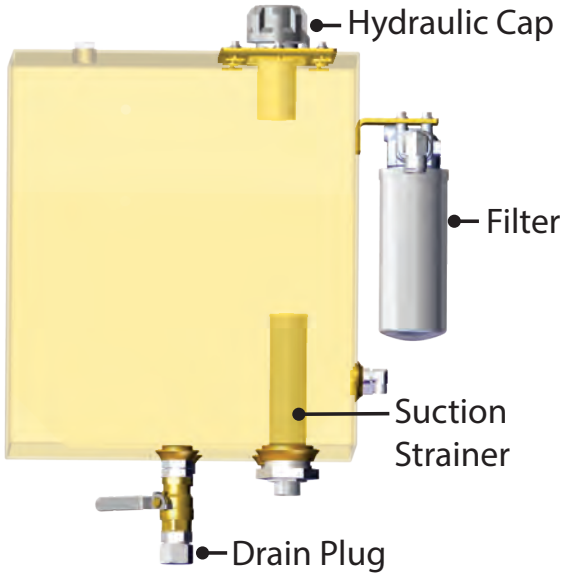


Figure 5-17. Hydraulic Suction Strainer

To change the hydraulic oil strainer:

1. Remove the hydraulic oil filler cap.
2. Remove the six (6) screws securing the strainer, then remove the strainer and gasket.
3. Install a new gasket, aligning the six screw holes in the gasket with the mounting holes on the reservoir.
4. Install the new strainer, aligning the holes in the strainer with the mounting holes of the gasket and secure the strainer with the six screws.
5. Fill the hydraulic oil reservoir with (recommend Citgo All Weather/All Temperature Multiviscosity hydraulic oil).

NOTICE DO NOT overfill the hydraulic oil reservoir with oil.

6. Check the oil level in the hydraulic oil reservoir. Add oil if needed.
7. Reinstall the hydraulic oil filler cap and tighten securely.

Priority Relief Valve

The priority flow divider in the hydraulic manifold is set to supply the power steering with approximately 4 GPM (15 LPM) of hydraulic flow. This flow goes to the steering circuit before any other circuit.

The Priority Relief Valve controls the maximum operating pressure for the power steering, brush lift and swing circuits. **(Figure 5-18)** This valve is located in the hydraulic pump attached to the rear of the hydrostatic pump on the engine flywheel.

These are signs the priority relief valve needs adjusting:

- Total or partial loss of steering functions or hard steering.
- Constant noise from the hydraulic pump when using steering or brush lift and swing.
- Hydraulic oil overheating.

⚠WARNING Only an experienced technician should service the hydraulic manifold (priority relief) as death or serious injury can result from improper valve adjustments. See your authorized LeeBoy/ Rosco dealer for assistance if experiencing steering function issues or hydraulic oil overheating.

To adjust the priority relief valve:

1. Set the park brake and ensure the transmission is in neutral. Use the foot brake as an extra precaution.
2. Plumb a 0 to 5000 PSI pressure gauge into the priority flow circuit.
3. Start the engine and allow the hydraulic oil to warm to at least 100° F.
4. Increase engine speed to 2500 RPMs.
5. Raise the brush until it stops, continuing to hold pressure to the lift cylinder until a gauge reading can be taken.
6. The pressure gauge should read 1500 +/- 50 PSI.
7. Adjust the relief pressure by removing the locknut and turning the adjusting screw:
   - Turn adjusting screw clockwise to increase pressure.
   - Turn adjusting screw counterclockwise to decrease pressure.

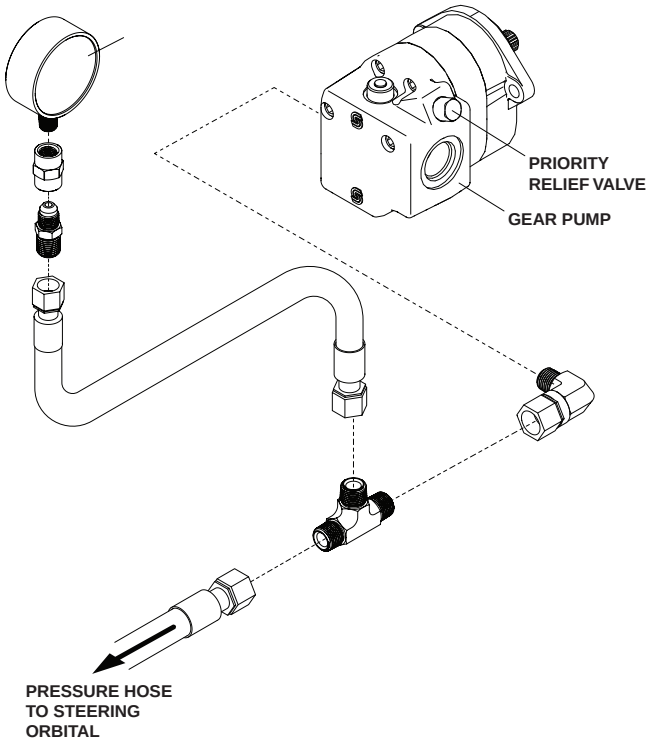


Figure 5-18. Hydraulic Manifold (Priority Relief)

Brush Drive Relief Valve

The brush drive circuit gets the remaining hydraulic flow after the priority divider supplies the power steering. The Brush Drive Relief Valve protects the brush drive circuit and is also located end of the rotation control valve under the cab on the right side of the broom. **(Figure 5-19)**

Adjust the Brush Drive Relief Valve if:

- Brush stalls frequently.
- Brush lacks sweeping power.
- There is a constant noise while sweeping.

⚠WARNING Only an experienced technician should perform this procedure as death or serious injury can result from improper valve adjustments. See your authorized Rosco/LeeBoy dealer for assistance if needed.

To adjust the Brush Drive Relief Valve:

1. Set the park brake and ensure the transmission is in neutral. Use the foot brake as an added precaution.
2. Plumb a 0 - 5000 PSI pressure gauge into the rotation flow circuit.
3. Turn the Brush Float control valve OFF by turning clockwise until stopped.
4. Start the engine and warm the hydraulic oil to at least 100° F.
5. Engage the brush drive control handle to turn brush in the forward direction.
6. Increase engine speed to 2500 RPMs.
7. Using the Brush Forward/Reverse control, turn the brush motor ON to engage the control valve.
8. The pressure gauge should read 2500 PSI +/- 100 PSI.
9. Disengage the brush drive control lever.
10. Adjust relief pressure by removing the Relief Pressure Locknut and turning the adjusting screw clockwise to increase pressure, or counterclockwise to decrease pressure.

NOTE: Turn the adjusting screw in 1/8 turn increments. Repeat turning increments until the correct pressure is obtained.

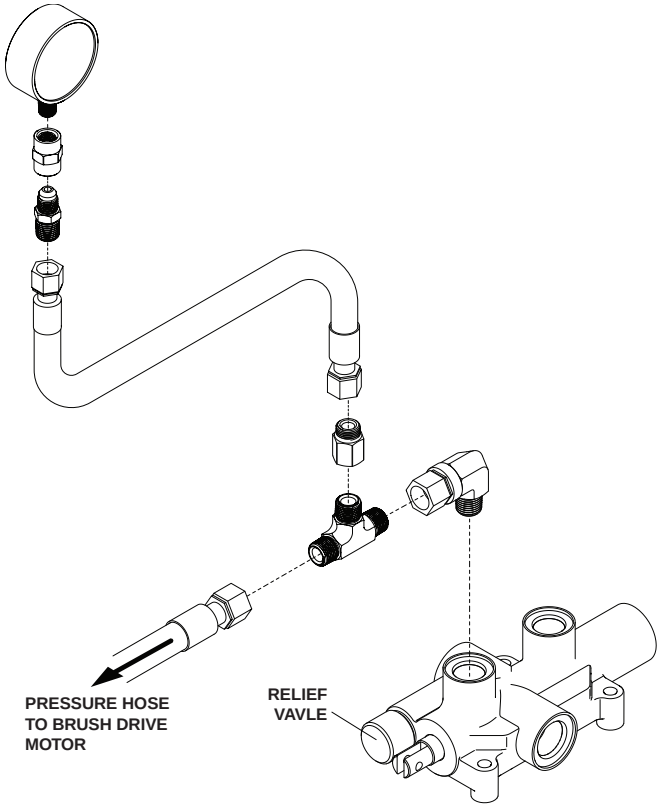


Figure 5-19. Hydraulic Manifold (Brush Drive)

CHASSIS MAINTENANCE

Tires and Wheels

Check the tires using an accurate tire gauge before operating your RB50 every day. Tires should be inflated to 80 PSI when tires are cold.

⚠CAUTION DO NOT exceed maximum inflation pressure for rims or wheels.

- Check tires for excessive wear, correct tire pressure and any damage regularly. ***(Refer to Table 5-3, Tire Wear Chart, on the following page.)***
- Use proper tire mounting equipment and experienced personnel for tire repair service.
- Check tires and wheels for cuts, bubbles, damaged rims or missing wheel bolts or nuts.
- Protect tires from contact with petroleum products and chemicals.
- Use proper tire mounting equipment and only allow experienced personnel to change tires.
- Carefully inspect any tire and rim assembly for damage before inflating tire. Contact a qualified repair service if needed.

NOTE: Use a tire pressure gauge with 1 PSI graduations.

NOTE: Limit air pressure to the proper level.

- It is extremely important to apply and maintain the proper torque to prevent loose wheels, broken studs and possible dangerous separation of wheels from the axle. Be sure to use only fasteners matched to the cone angle of your wheel. **ALWAYS use a torque wrench.**

To check tire air pressure:

1. Install lock on air chuck or stand-off gauge on tire stem. **(Figure 5-20)**
2. Turn on air supply. Stand to one side of the tire during inflation.
3. After proper inflation (the recommended pressure is 0 PSI), release chuck or stand-off gauge.
4. Torque wheel lug nuts to 85 ft lb.

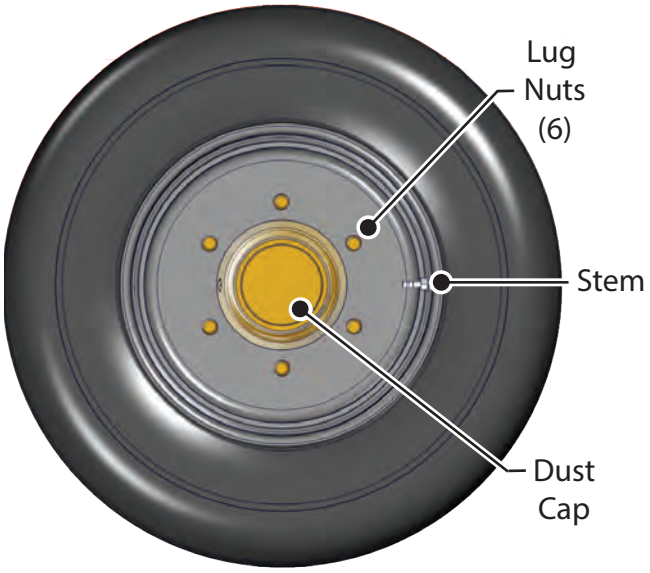


Figure 5-20. Tire

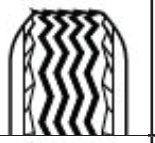
To inflate tires:

1. Install lock on air chuck or stand-off gauge on tire stem. **(Figure 5-20)**
2. Turn on air supply. Stand to one side of the tire during inflation.
3. After proper inflation (the recommended pressure is 0 PSI), release chuck or stand-off gauge.
4. Torque wheel lug nuts to 85 ft lb.
   - Protect tires from contact with petroleum products and chemicals.
   - Repair or replace damaged tires. DO NOT operate the machine with damaged tires.
   - Visually inspect the wheels for damage prior to operating the broom. Replace bent or cracked wheels immediately.

⚠DANGER Explosive separation of a tire and rim parts can cause serious injury or death.

⚠WARNING Never cut or weld on inflated tire/rim assembly. Heat from welding can cause an increase in pressure and may result in tire explosion.

Table 5-3.Tire Wear Chart

WEAR PATTERN		CAUSE	ACTION
	Center Wear	Over Inflation	Deflate tire to correct load level.
	Edge Wear	Under Inflation	Inflate tire to correct load level.
	Side Wear	Loss of camber or overloading.	Get align-ment. Ensure load doesn't exceed axle rating.
	Toe Wear	Incorrect toe-in.	Get alignment service.
	Cupping	Wheels out of balance.	Check balance and bearing adjustment.
	Flat Spots	Wheels locked and tire skidding.	Adjust brakes.

Change Tires

⚠WARNING It is recommended that you use a hydraulic lift if available when raising the vehicle. Otherwise, use jack stands on a solid, flat surface to prevent the machine from tipping when it is raised. Block the remaining wheels using sturdy wheel chocks. DO NOT lie under the machine while supported only by a jack. Death or serious injury can result from improperly supporting a raised machine.

To replace the tires:

1. Loosen lug nuts.
2. Release any air that remains in the tire.
3. Use a jack or hoist to lift the side of the machine.
4. Place supports under the machine.
5. Attach a sling or other support to the tire.
6. Remove the six (6) lug nuts from the wheel.
7. Release any air that remains in the tire.
8. Using standard tire removal tools, remove the tire from the wheel.
9. Install new tire on the wheel.
10. Measure PSI using tire pressure gauge and inflate to the proper PSI.
11. Align the mounting holes with the studs on the hub.
12. Reinstall lug nuts (6), tightening every other lug nut alternately in a star pattern.
13. Torque to 80 ft lbs.
14. Remove supports and lower jack.

Repack Front Wheel Bearings

The wheel bearings secure the wheel and tire to the vehicle on the drive shaft inside the wheel hub. While wheel bearings are designed to last 100,000 miles, they can lose lubricity over time. Dirt and debris can also seep into the wheel bearings. If the wheel bearings aren't cleaned and repacked, they can wear out and possibly break or damage other wheel components.

When any of the following symptoms appear, the wheel bearing is damaged and should be replaced on ***both*** wheels of the same axle.

- Abnormal and excessive tire wear on the inside or outside edge of tire.
- Grinding or roaring noise from a tire. This symptom indicates metal-on-metal contact and definitely requires wheel bearing replacement.
- The steering wheel bounces and vibrates at lower speeds, vibrating more as the machine accelerates.

NOTICE **Cleaning and repacking or replacing wheel bearings is a messy job. Wear gloves and have plenty of shop rags and room to work around the entire wheel hub. Dispose of used grease in accordance with environmental law.**

⚠WARNING If using compressed air for drying wheel hub components after cleaning, wear eye protection! Compressed air can cause serious personal injury if misused. Use compressed air at safe low levels. NEVER point the nozzle at others.

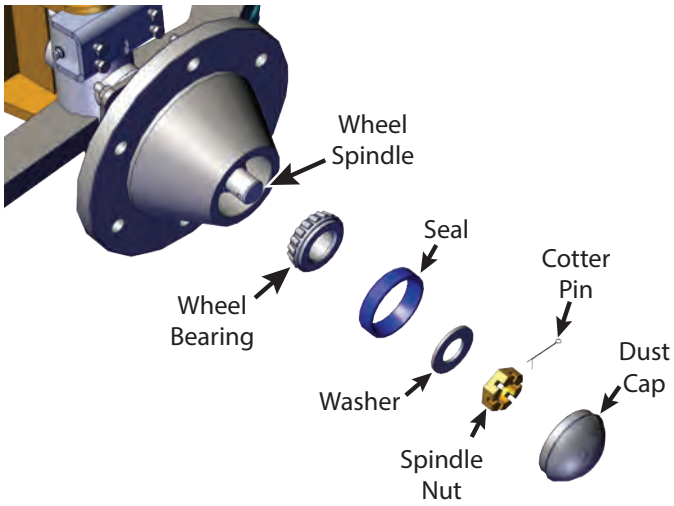


Figure 5-21. Wheel Hub Components

To clean and repack (or replace) wheel bearings:

1. Park the broom on a level surface in a designated area where grease does not touch or run off onto the ground.
2. Using a hydraulic lift (if possible) or floor jack, raise vehicle with a pair of jack stands on the front axle to service wheel bearings. Use wheel chocks on the other wheels for safety.
3. Remove dust cover and lug nuts from one wheel, then remove tire and wheel hub, placing it clearly away from the work area. **(Figure 5-20 *on Page 5-22)***
4. Remove cotter pin to release spindle nut, then remove spindle nut and washer. **(Figure 5-21)**
5. Carefully pull the wheel bearing and seal off the axle spindle. Inspect wheel bearing and seal for wear or damage.
   - Discard bearing if replacement is needed. It is recommended that you discard the seals and replace with new ones.
   - Otherwise, thoroughly clean wheel bearings, wheel hub, axle spindle and mounting hardware with solvent. Once the excess grease is removed, you can spray liberal amounts of brake cleaner inside the wheel bearings and wheel hub for additional cleaning.
6. Work fresh grease into the wheel bearings. You can put the wheel bearing and grease into a plastic zip lock bag to really work the grease into every small wheel and bearing.
7. Work fresh grease inside the wheel hub, ensuring the outer edges are fully covered, and put liberal amounts of grease onto the axle spindle.
8. Reinsert the wheel bearing, ensuring it is oriented as shown in **Figure 5-20**, then new seal, washer, spindle nut. Continue packing grease inside all the parts as you install them.
9. Tighten the spindle nut until the wheel "drags" during hand rotation.
10. Back off the nut 1/8th of a turn and reinsert cotter pin.

- You can spin the hub and rotor to test for noise and smoothness before reinstalling the wheel and tire. When packed correctly, the rotor spins freely without noise.

11. Reinstall tire and wheel. Torque lug nuts. (***See Section 2, Torque Specifications)***
12. Repeat procedure on the other wheel.

Change Drive Axle Oil

Change the drive axle oil every year. Check the oil level and fill if needed quarterly.

To fill the oil:

1. Rotate the hub on one side of the front axle until the plug is in the 12 o'clock position.
2. Remove the plug. **(Figure 5-22)**
3. Add one quart of 80W-90LS GL5 oil.
4. Replace the plug.
5. Follow Steps 1 - 4 on the other side.

To change the drive axle oil:

1. Remove the level plug in the center of the housing.
2. Remove the breather plug in the housing.
3. Add oil until it starts coming out of the level plug.
4. Replace the level plug.
5. Allow the axle to sit level for at least five minutes to allow the oil to flow to the hubs.
6. Remove the level plug and check oil level.
7. Add more oil if needed.

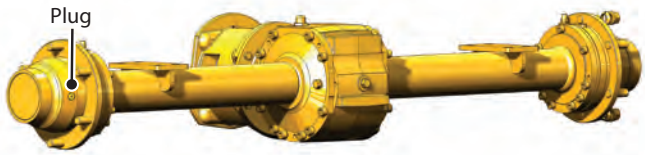


Figure 5-22. Front Drive Axle

Park Brake Adjustment

NOTICE **Only a qualified service technician should adjust the parking brake as serious damage can occur to this important component. See your authorized LeeBoy/Rosco dealer for assistance.**

⚠WARNING **Serious injury or even death can result from an improper park brake adjustment.**

Only a qualified technician should perform the following procedure:

1. Loosen the park brake bolt.
2. Rotate the park brake knob clockwise to tighten the brake cable or counterclockwise to loosen the brake cable.
3. When the desired tension is reached, retighten the park brake bolt.

Brakes

Check the master cylinder fluid daily. The master cylinder fluid filler cap is located on the cab floor behind the brake pedal. **(Figure 5-23)** Only fill one inch below the top of the fill neck with All Weather All Temperature VG32.

NOTICE **If the master cylinder fluid is frequently low, check the hydraulic system for leaks.**



Figure 5-23. Brake Fluid Fill

5

DUST SUPPRESSION SYSTEM (OPTION)

Clean the water system strainer after each use. Simply unscrew the strainer from its mount under the water tank at the rear of the vehicle, clean, and reinsert. Remove the drain plug to drain the water tank. **(Figure 5-24)**

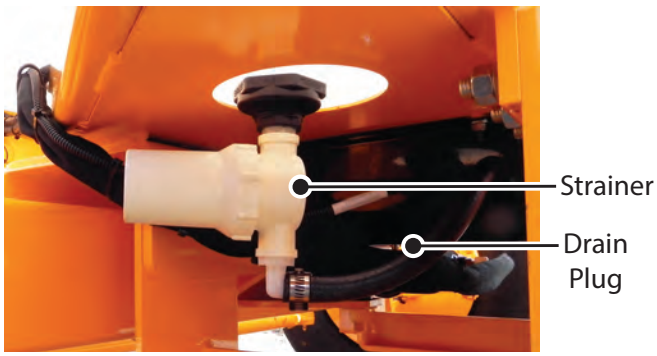


Figure 5-24. Water Strainer and Drain Plug

Inspect the brush water nozzles daily to ensure they are free from clogs and operating properly. If a nozzle becomes clogged:

1. Remove the nozzle by unscrewing it from the spraybar.
2. Clean the nozzle slots, being careful not to damage the nozzle opening as the spray pattern will be adversely affected.
3. Reinsert or replace the nozzle if needed.
4. If the nozzle is still clogged or not spraying properly, contact your authorized Rosco/LeeBoy dealer for a replacement nozzle(s).
   - Flush the spraybar line and nozzles at the end of each operating season (or more often for extended use).
   - Ensure the water tank, strainer, pump and all lines are completely drained to prevent damage from freezing if storing broom in temperatures below freezing.

COMPONENT REPLACEMENT

Change Brush Core

1. Park broom on flat surface, place machine in neutral, turn off the engine and set park brake.
2. Remove both cotter pins from brush cover. **(Figure 5-25)**

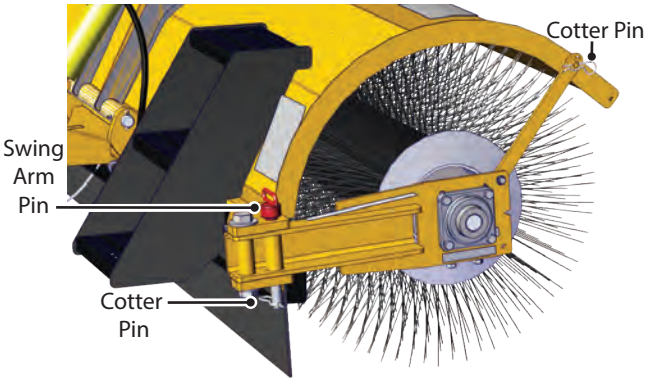


Figure 5-25. Brush Core Pins

3. Remove the swing arm pin.
4. Pull the brush core away from the drive hub and slide out. Set aside.
5. Check hub on both sides for damage. (Consult your authorized dealer if any damage is noted.)
6. Grease the U-Joint on the idle hub.
7. Slide new brush onto the slotted drive hub ensuring the key stock inside the brush core goes onto the drive hub side.
8. Align the idle hub with the smooth end of the brush core and push the swing arm back into position into the brush core, keeping hands away from pinch points. Carefully guide the hub by hand if necessary to start into the core center.

NOTE: Brush carriage may need to be raised slightly to install a new core if the old brush core was worn.

9. Once the idle hub is fully engaged, reinsert the swing arm pin.
10. Reinsert swing arm pin and cotter pins on brush cover.

Replace Batteries

⚠WARNING Batteries create explosive gases when charging. DO NOT expose these gases to sparks or open flame during maintenance. Always perform battery service in a well-ventilated area or outside. Serious injury can occur from exploding batteries.

1. Remove the black ground (-) cable from the negative (-) battery terminal, followed by the red (+) cable from the positive (+) battery terminal. **(Figure 5-26)**

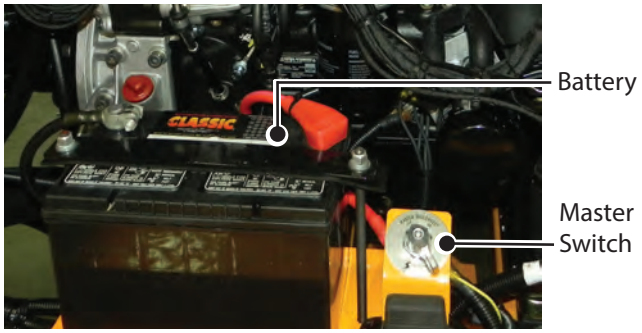


Figure 5-26. Battery System

⚠WARNING If the negative cable is grounded during removal, sparks will not occur. If the positive cable is grounded during removal, sparks will occur and could ignite explosive gases.

2. Remove the hold down bar.
3. Inspect the batteries for cracks, wear and damage that could result in leaks.
4. Carefully remove the batteries from the battery plate, noting battery position during removal.

⚠WARNING DO NOT dry batteries using compressed air. Allow the batteries and surrounding area to dry. Battery acid can cause severe injury if blown into the face or other areas of the body. If injury occurs, seek medical attention immediately. Flush eyes and skin for 30 minutes with clean water.

5. Make sure the battery cells are sealed. Clean the batteries, battery tray, cables, and brackets with baking soda and warm water. Allow to air dry.
6. Place the batteries in proper position in the battery tray.
7. Install hold down bar. Tighten bar to retain battery, but not so tight as to damage battery.

⚠WARNING When installing batteries, always install the positive cable first followed by the negative cable. Reversed polarity can damage the electrical system.

8. Install red (+) battery cable to the positive (+) battery terminal.
9. Install the black (-) battery cable to the negative (-) battery terminal.
10. Reattach the battery box cover and reconnect the battery box locks.

Replace Radiator Hoses

Radiator hoses should be replaced if worn or frayed. Replacement parts are listed in **Section 7, Illustrated Parts List (IPL).**

⚠WARNING **This procedure should only be performed when the equipment is not operating.**

⚠WARNING **DO NOT remove the radiator cap while the engine is still hot. The radiator is under pressure. Removing the cap could cause serious injury.**

1. Turn off ignition, remove key, and allow the engine to cool down before proceeding:
2. Remove the radiator surge tank cap.
3. Drain radiator into a suitable container.
4. To remove upper hose, loosen the two hose clamps and disconnect. If clamps are not damaged, retain for installation.
5. To remove lower hose, loosen the two hose clamps and disconnect. If clamps are not damaged, retain for installation.
6. Install new hoses in reverse order of procedures above.

WINTERIZING FOR STORAGE

Perform the following procedures to prepare the machine for winterizing and storage.

1. Store the machine in a dry, protected area:
   - Cover seat with plastic. Place a “DO NOT OPERATE” tag on the main control panel.
   - Cover all exposed control panels with tarps or plastic.
   - Ensure all cab doors and windows are closed and seal properly.
   - Ensure the tires are properly inflated.
   - Replace worn or damaged decals if needed.
2. Inspect the machine thoroughly and repair worn or damaged parts. Inspect all hoses, couplers, fittings and cylinders. Tighten loose fittings and replace any worn hoses found.
3. Lubricate all grease points:
   - Coat all exposed cylinder shafts with lithium-based grease.
   - Ensure all exposed threads are lubricated.
   - Ensure cables are lubricated and move freely.
   - If the machine has high- and low-speed chains, be sure they are oiled.
4. Check air filter. Clean or replace air filter elements if needed.
5. Check engine oil level and fill if needed.
   - Drain dirty engine oil, fill with new oil and run the engine five minutes to allow the oil to penetrate components.

⚠WARNING **To avoid the danger of exhaust fume poisoning, DO NOT operate the engine in a closed building without proper ventilation. Check hydraulic oil sight gauge and fill if needed.**

NOTICE **Hydraulic oil that is contaminated must be drained, the filter elements replaced and the hydraulic tank refilled. Failure to perform this maintenance procedure if needed could result in premature failure of the pumps or motors.**

6. Ensure the fuel tank is full to prevent condensation in the fuel tank. Add cold-weather additive when appropriate.
7. If machine is equipped with Diesel Exhaust Fluid (DEF), ensure the DEF tank is full.

NOTICE **IF machine is equipped with DEF and is stored for more than one year, replace the DEF fluid before operating to prevent SCR system malfunction and/or damage.**

8. If machine is equipped with a citrus tank, ensure the tank is full to prevent rust.
9. If machine is equipped with water tank(s), the tank should be drained. The pump and filter must also be purged to prevent freezing.
10. If machine is equipped with windshield washer bottle, ensure the bottle is full of window washer fluid with a cold-weather blend.
11. Perform specific gravity test on engine coolant. Drain and replace or fill with antifreeze mixture to prevent freeze damage if needed.

KUBOTA ENGINE NOTICE: If not using anti-freeze, open the cock at the bottom of radiator and remove the pressure cap to drain water completely. Leave the cock open. Place a "No Water" sign on the pressure cap.

12. Disconnect battery, adjust the electrolyte level, and recharge it. Store battery inside in a cool, dry place. Ensure breakers are in the OFF position.
13. Remove steering control boxes and electrical cords. Store inside in a cool, dry place.
    - If equipped with any plugged-in remote sensors (i.e., sonic auger sensor), disconnect, remove and store inside in a cool dry place.
14. Cover the exhaust and intake filter. Be sure to seal the exhaust cover with tape to prevent moisture from entering, especially if storing outside.
15. If the machine is equipped with a propane system, remove the propane gas bottle and store in a safe area away from the machine. Purge the gas system and seal connection(s).
16. Lower machine components and attachments (i.e., screed, broom, blade, etc.) to the lowest position, resting on blocks to prevent ground contact or pavement damage.
17. Clean the machine, including the undercarriage. Paint chipped or rusty areas to prevent further rusting. Lock all lockable panels, cab doors and spare tire. Remove ignition key and store keys in a safe place.
18. If storing machine for an extended time, run the engine five minutes every two to three months (under no load) to keep engine free from rust caused by moisture in the air. If you do not run the engine for six months or longer, apply engine oil to the valve guide and stem seal before starting.

Periodic Maintenance in Storage

If storing the Rosco RB50 more than two months, perform the routine maintenance procedures for every 100 hours.

Also be sure to:

- Keep battery fully charged and check the electrolyte level regularly.
- Check the hydraulic fluid on a regular basis for possible moisture contamination. This is especially important in climates with varying temperatures and humidity as condensation can develop inside the hydraulic tank.

NOTICE **Hydraulic oil that is contaminated must be drained, the filter elements replaced and the hydraulic tank refilled. Failure to perform this maintenance procedure if needed could result in premature failure of the hydraulic pumps or motors.**

- Reinstall the battery to start the engine and allow the engine to warm until all machine functions are at operating temperature at least twice per month.
- Continue to perform routine maintenance procedures as required using the Periodic Maintenance Table on ***Page 5-3.***

Removing from Storage

1. Perform maintenance procedures described above.
2. Check all fluid levels, belt tensions, and bolt torques.
3. Replace battery.
4. Replace alternator belt.
5. Clean any excess grease or rust inhibitor from all exposed cylinder rods, seals and O-rings.
6. Disengage hydraulic brush lift lock valve.
7. Start engine and allow machine to warm to operating before driving out of storage.



Figure 5-27. Welding

WELDMENT

If the machine has been damaged or metal parts worn due to extensive use, minor welding may be necessary for repair. Welding is very dangerous. Only individuals who are properly trained should weld in a dedicated area away from others as even looking at ultraviolet light from the flame can cause serious eye injury.

Welding hazards include toxic fumes, harmful dust, light radiation, burns, combustion from sparks and the potential for extensive damage to the eye. There is also a danger of electric shock. If combustible or flammable materials are nearby, the heat and sparks produced by welding can cause fires or explosions.

DISCONNECT BATTERY TERMINALS BEFORE WELDING ON THE MACHINE.

Electrical current can ignite fire or cause an explosion, therefore it is critical to turn off power to the machine. It is also important to grind off the paint to ensure proper ground on the area being welded.

DO NOT MAKE MODIFICATIONS TO THE MACHINE. ONLY weld if needed for repair purposes. Weldment modifications to the machine can void your warranty.

Before welding on the machine:

PREPARE:

- Clear the area of any fire hazards, flammable liquids and solids, or any items that could catch on fire.
- Ensure a fire extinguisher is available in the work area.
- Hang welding blanket(s) around and underneath the work area.
- Paint and rust should be removed from the welding area to prevent arcing in other areas or components. An indirect ground can cause arcing that will damage machine components. Only connect the ground to the component being welded. DO NOT connect the ground to other bolted assemblies, wheels, bearings, motors, tracks, torque hubs, etc.

DISCONNECT:

- Turn the master battery switch to the OFF position.
- Disconnect the battery positive and negative terminals.
- Disconnect engine ECU(s) controller (if equipped).
- Disconnect engine DEF controller (if equipped).

TROUBLESHOOTING

Hydraulic Motor Installation Start-Up Procedure

Pre-Start

If the hydraulic system is down as a result of a major component failure:

1. Drain and clean the tank and system components (hoses, valves, fittings and cooler) to ensure it is free from metallic debris and other contamination. Failure to do so may result in damage to the pump(s) and/or other components on start-up.
2. Change all the filters.
3. Change the fluid. On large systems where the cost of changing the fluid may be prohibitive, the fluid should be flushed until a cleanliness level of ISO 4406 18/16/13 or better is achieved.

Installation and Start-Up

When installing hydraulic motors, it is important that the mounting flange of the motor makes full contact with the mounting surface of the application. Mounting hardware of the appropriate grade and size must be used:

1. Use Grade 8 socket-head capscrews to attach the motor.
2. Install lock washer over capscrew.
3. Apply Loctite 243 (blue) to the capscrew threads.
4. Install extra thick, hardened H-D thick washers over the capscrew.
5. Install the capscrew-hardened washer and lock washer in the SAE two-bolt flange.
6. Torque the capscrew to the wet torque value defined for the capscrew grade and size.

NOTICE **It is CRITICAL to use the correct mounting hardware.**

Hubs, pulleys, sprockets and couplings must be properly aligned to avoid inducing excessive thrust or radial loads. Although the output device must fit the shaft tightly, a hammer should never be used to install any type of output device onto the shaft. The port plugs should only be removed from the motor when the system connections are ready to be made.

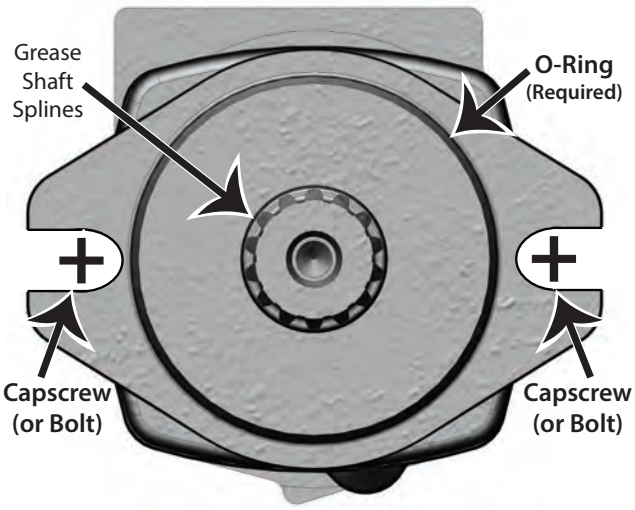


Figure 5-28. Hydraulic Motor Mount

To avoid contamination:

- Remove all matter from around the ports of the motor and the threads of fittings.
- Fill the case of the motor being used (piston-type, gear motor, gerotor motor, geroller motor or vane motor) with clean hydraulic fluid through the highest case drain port and connect the case drain line. Failure to do so will result in damage to the motor through inadequate lubrication on start-up.
- Units that are mounted vertically with the shaft up require special attention to ensure the fluid level in the case is high enough to lubricate the front shaft bearing(s).

Once all system connections are made, the motor should be run for 15 - 30 minutes at no-load and half-speed to bleed air from the hydraulic system.

Troubleshooting Chart

The troubleshooting chart below identifies the most common symptoms of mechanical problems. Fault codes are also shown on the PV380 Digital Display unit on the Operator Control Panel.

If the problem persists, contact your authorized LeeBoy dealer for assistance or repair.

Table 5-4. RB50 Broom Troubleshooting

SYMPTOM	CAUSE	REMEDY
Engine difficult to start or does not start.	Defective battery or low battery charge.	Replace or charge battery as applicable.
	Insufficient fuel supply.	Fill fuel tank.
	Safety switch is faulty.	Replace.
	Solenoid doesn't function properly.	Ensure wires are tight. Clean solenoid plunger. Replace solenoid coil. Replace or rebuild starter or solenoid.
	Switch box unplugged.	Replug switch box.
	Faulty starter.	Repair or replace.
	Engine cranking speed too low.	Clean battery terminal connections. Charge or replace battery.
	Insufficient fuel supply to injector nozzles.	Check fuel system. Clean or replace fuel filter.
	Air in the fuel system.	Check for air leaks in the low-pressure side of the fuel system. Prime fuel system. Refer to your Engine Operator's Manual.
	Fuel supply contaminated.	Verify by operating engine with a known fuel quality.
Engine turns over but does not start.	Low fuel.	Add fuel to fuel tank.
	No voltage to fuel shut-off solenoid.	Voltage should be 9 volts with the ignition switch in the ON or START position.
	No fuel to injector pump.	Check fuel supply system.
	Fuel filter plugged or restricted.	Replace fuel filter.
	Intake or exhaust system restricted.	Check for and remove restrictions.
Engine doesn't shut off.	Electrical wiring fault supplying power to the fuel solenoid when key is in the OFF position.	Repair wiring.
	Faulty diode in engine wire harness at alternator.	Check diode wire connection and/or replace diode.
	Injection pump fuel solenoid inoperative.	Check solenoid for defects or foreign material inhibiting proper operations.
	Engine operating on fumes drawn into air intake.	Locate and isolate source of fumes.
	Low idle set too high.	Set idle to specifications.